



Simulation of digital lensless holographic microscopy holograms: a physics-image processing approach

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Abstract: This work presents a method for simulating digital lensless holographic microscopy (DLHM) holograms using a physics-based image processing approach. While DLHM has gained significant attention in biology, biomedicine, and environmental monitoring, the current modeling of DLHM holograms has been limited, hindering potential applications, including learning-based solutions and generative model training. In this study, the DLHM propagation process is decomposed into the diffraction of a complex-valued spherical wavefront and the non-homogeneous magnification of the diffracted field that encodes the sample information, which accelerates and enhances the hologram simulation. The proposed model is validated by comparing simulated and experimental holograms of standard test targets under diverse imaging conditions. Comparative analyses are conducted against other DLHM hologram modeling methods, including direct Rayleigh-Sommerfeld diffraction, its convolutional implementation, and the Fresnel-Bluestein formalism. The proposed model is shown to outperform these methods in overall similarity to experimental recordings across a wide range of imaging conditions while maintaining computational efficiency. This DLHM hologram modeling approach provides researchers with a powerful tool for simulating trustable holograms. The model can be publicly accessed through the open-access repository <https://github.com/mloper23/DLHM-model>.

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1. Introduction

Digital Lensless Holographic Microscopy (DLHM) is perhaps the simplest microscopy methodology as it only requires a spherical wavefront light source, a sample, and a sensor to image transparent or partially opaque samples [1]. Owing to these features, DLHM has found extensive applications in diverse fields such as biology, biomedicine [2], and environmental sciences [3], offering valuable features like particle tracking [4] and specimen classification, and, with minor hardware modifications, DLHM can provide polarization sensitivity measurements [5]. However, the spherical nature of the coherent illumination in the technique imposes challenges when simulating such experiment [6–8]. The classical approach involves computing the non-approximated version of the Rayleigh-Sommerfeld propagation integral [9]. However, the computational complexity of this implementation discourages its extensive utilization [6]. The substantial computational burden imposed by this approach limits its practicality and renders it non-attractive for numerous applications. While general numerical propagation approaches based on the fast Fourier transform (FFT) can reduce the computational complexity of non-approximated discretized diffraction integrals in DLHM hologram modeling, they are primarily designed for

plane wavefront illumination, as the discretization on both planes must remain constant [10]. When a diverging spherical complex-valued wavefield is included, its discretization on the sample plane becomes a prevailing issue. Notable approaches include the classical angular spectrum method [10], its version with adapted sampling rate [11], and the convolutional version of the Rayleigh-Sommerfeld formula [8]. Additionally, methods like the Fresnel-Bluestein transform [12] incorporate spherical wavefronts by using a non-linear discretization to address the sampling issues. However, since this method is based on Fresnel propagation, it fails for low magnifications or for numerical apertures greater than 0.58, both of which imply short propagation distances [12]. Diverse approaches have emerged, such as the homeomorphic Fourier Transform [13], the non-uniform discretization with compact space bandwidth [14] and even the alternative of using the Whittaker cardinal basis to avoid the discretization issue of spherical wavefront [15]. Overall, classical DLHM modeling approaches face significant limitations due to their inability to accurately produce high numerical aperture and high magnification synthetic DLHM holograms efficiently, where the intrinsic free-space magnification of the diffracted wavefield must be reproduced [11]. Further advancements in developing accurate DLHM hologram modeling methods could significantly enhance the understanding of the technique's operational principles, limitations, and the behavior of the recorded holographic data. Specifically, the modeling approach could pave the way for developing learning-based DLHM methods with improved accuracy [16], reducing the heavy dependence on vast amounts of experimental data [17]. DLHM can be leveraged even more effectively by refining the modeling techniques, which is crucial for optimizing its performance, improving accuracy, and advancing its applications.

This work presents an innovative approach for modeling DLHM holograms by decomposing the involved physical phenomena into independently modeled components. This method results in a more accurate, realistic, and robust simulation tool for DLHM than previously reported. The diffraction of light using a spherical wavefront source is examined, involving two principal phenomena: the diffraction of a complex-valued spherical wavefront after illuminating the sample and the magnification of this diffracted field. By independently computing these effects, a comprehensive understanding of the underlying physics is attained, enhancing the proposed model. This method is contrasted with classical approaches by initially validating the ranges of numerical apertures and magnifications where each simulation method works, considering the discretization and sensor size of commercially available sensors. The performance of the method is evaluated by comparing the achieved spatial resolution in the reconstructions of both experimental and simulated holograms, recorded and modelled, respectively, with identical DLHM optical parameters. For the validations, a calibrated phase target with the USAF 1951 test layout and a Siemens Star target are used to measure the lateral resolution.

The first part of this manuscript briefly describes the classical simulation formalisms used in DLHM, followed by the methodology of the proposed approach, and the implemented algorithm. Finally, the comparison of the methods is described, to finish with the validation of the model and conclusions.

2. Digital lensless holographic microscopy

The physical model of in-line Gabor's holography involves the diffraction of an illuminating wavefront through a sample. When adapted to light waves, Gabor's original concept for electron microscopy [18], can be implemented straightforwardly by employing a collimated beam to illuminate the sample of interest. The resulting diffracted beam is then recorded on a subsequent plane. In this configuration, the amplitude and phase of the illuminating wavefront remain constant throughout propagation. Consequently, after interacting with the sample, the wavefront only evolves according to the object's characteristics, and no magnification of the diffraction pattern is achieved in this case. In contrast, when a spherical wavefront illumination is utilized, the resulting diffracted pattern undergoes magnification during propagation [1]. This feature is

the working principle behind Digital Lensless Holographic Microscopy (DLHM), where point sources produce spherical wavefront illumination [19]. In this technique, the closer the sample is to the point source, the greater the magnification is at the recording plane [1]. Nonetheless, to achieve high-resolution reconstructions in DLHM [20], it is important to recognize that the resolution is primarily limited by diffraction and sampling criteria [21,22]. As a result, DLHM offers the advantage of utilizing cost-effective sensors and employing more straightforward numerical reconstruction methods than other lens-free in-line imaging modalities that achieve comparable lateral resolution; besides, they typically need additional operations, such as multi-shot recordings, deep learning processing, or super-resolution techniques [23]. Figure 1 depicts a schematic representation encompassing the key elements to consider in DLHM modeling.

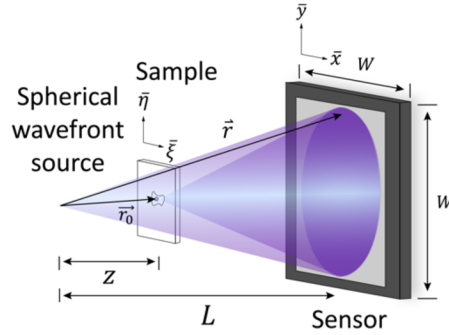


Fig. 1. Scheme of the DLHM setup.

In DLHM, the sample $S(\vec{r}_0)$, located at a distance z from the point source, is illuminated by a diverging spherical wavefront $PS(r_0)$ given by $\frac{\exp[ik \cdot \vec{r}_0]}{|\vec{r}_0|}$. The transmitted wavefield is magnified by free-space propagation until it reaches a digital sensor located at a distance L from the point source. The mathematical description of the optical wavefield of the diffraction pattern at the sensor plane can be modeled using the Rayleigh-Sommerfeld diffraction formula [6], as presented in Eq. (1):

$$U(\vec{r}) = \int_{\text{sample}} S(\vec{r}_0) \frac{\exp[i\vec{k} \cdot \vec{r}_0]}{|\vec{r}_0|} \frac{\exp[i\vec{k} \cdot (\vec{r} - \vec{r}_0)]}{|\vec{r} - \vec{r}_0|} d\vec{r}_0 \quad (1)$$

where the position vectors $\vec{r}_0 = (\xi, \eta, z)$ and $\vec{r} = (x, y, L)$ denote the sample and sensor planes, respectively. The recorded information at the sensor plane, the DLHM hologram $H(\vec{r})$, is equal to the squared magnitude of this optical wavefield

$$H(\vec{r}) = U(\vec{r}) U^*(\vec{r}). \quad (2)$$

3. Conventional numerical modeling approaches for DLHM

Computationally modeling a hologram inherently involves a discretization process of the input (sample), output (sensor) planes, and the propagation kernel, which naturally imposes sampling constraints as presented by [24–26]. Generally, the discretization process begins by considering the continuous spatial coordinates at the sample plane $\vec{r}_0 : (\xi, \eta, z)$ as a discrete rectangular grid of $M \times N$ pixels and with pixel size $\Delta\xi \times \Delta\eta$. The discrete coordinates of this plane are represented by $\xi = m\Delta\xi$ and $\eta = n\Delta\eta$, where m and n are integer numbers ranging from $0 \leq m \leq M-1$ and $0 \leq n \leq N-1$. A similar process occurs on the sensor's plane, which is represented by a rectangular grid of $P \times Q$ pixels, each of size $\Delta x \times \Delta y$. The coordinates in this plane are represented by $x = p\Delta x$ and $y = q\Delta y$, where p and q are integer numbers ranging from $0 \leq p \leq P-1$ and $0 \leq q \leq Q-1$. A visualization of the discretization process on DLHM is displayed on Fig. 2.

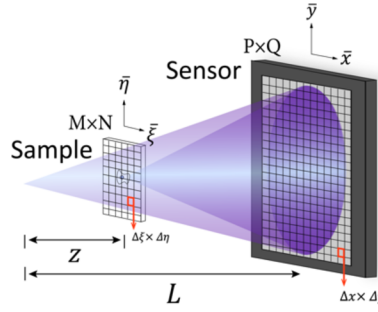


Fig. 2. Scheme of the DLHM setup, with discretized coordinates.

3.1. Discrete Rayleigh-Sommerfeld integral

The direct discretization of the Rayleigh-Sommerfeld propagation integral presented in Eq. (1) involves replacing the double integral by a double summation and substituting the vectorial operations by their corresponding discretized values [9]

$$U(p\Delta x, q\Delta y, L) = -\frac{1}{2\pi} \sum_{m=1}^M \sum_{n=1}^N S(m\Delta\xi, n\Delta\eta, z) \frac{\exp[ik|r_0|]}{r_0} \left(ik + \frac{1}{r_{01}} \right) \frac{\exp[ik|r_{01}|]}{r_{01}} \frac{L-z}{r_{01}} \cdot \Delta\xi\Delta\eta \quad (3)$$

where $r_{01} = \sqrt{(p\Delta x - m\Delta\xi)^2 + (q\Delta y - n\Delta\eta)^2 + (L-z)^2}$ and $r_0 = \sqrt{(m\Delta\xi)^2 + (n\Delta\eta)^2 + z^2}$. This process requires a total of $(M \times N \times P \times Q)$ complex-valued operations, which is impractically large for practical uses [9] and hinders the use of the discrete Rayleigh-Sommerfeld in modeling DLHM holograms.

3.2. Convolutional Rayleigh-Sommerfeld integral

By considering the explicit form of the distance r_{01} as displayed in Eq. (3), the Rayleigh-Sommerfeld integral is a linear space-invariant operation [9]. Hence, the convolution theorem can be used to compute the propagated field using Fourier transforms ($\mathcal{F}$) [10]. Therefore, its discretization is given by:

$$U(p\Delta x, q\Delta y, L) = -\frac{1}{2\pi} \mathcal{F}^{-1} \left\{ \mathcal{F} \left\{ S(m\Delta\xi, n\Delta\eta, z) \frac{\exp[ik|r_0|]}{r_0} \right\} \cdot \mathcal{F} \left\{ \left(ik + \frac{1}{R} \right) \frac{\exp[ik|R|]}{R} \frac{L-z}{R} \right\} \right\} \quad (4)$$

where $R = \sqrt{(p\Delta x)^2 + (q\Delta y)^2 + (L-z)^2}$. It is important to note that the implementation of Fourier transforms as FFTs requires that the input and output planes have the same number of pixels, therefore $M = P$ and $N = Q$ must be set accordingly, and for the convolutional approach the pixel sizes must also be identical, limiting the achievable magnification to unitary.

3.3. Fresnel-Bluestein transform

While the convolutional Rayleigh-Sommerfeld method effectively eases the computational time, it imposes the requirement of maintaining the same discretization on both the input and output planes. This constraint can lead to issues with correctly discretizing the samples, particularly

for high numerical apertures, where variations between the sample and sensor planes are more pronounced. To address these limitations, Bluestein's substitution of the discrete Fourier transform kernel [27] is introduced, allowing for independent sampling of the input and output planes. This modification, initially introduced for DLHM reconstruction by Kreuzer [28,29], is adapted to further account for the simulation of high numerical apertures by expanding the position vector to second-order approximation as used by [12]

$$r_{01} = \sqrt{(p\Delta x - m\Delta\xi)^2 + (q\Delta y - n\Delta\eta)^2 + (L - z)^2} \approx R \left(1 + \frac{1}{2} \frac{(m\Delta\xi)^2 + (n\Delta\eta)^2}{R^2} - \frac{m\Delta\xi p\Delta x + n\Delta\eta q\Delta y}{R^2} \right) \quad (5)$$

with $R = \sqrt{(p\Delta x)^2 + (q\Delta y)^2 + (L - z)^2}$. Considering this approximation, it is assumed that the dimensions of the recording plane are much larger than those of the sample plane. Since R depends on a quadratic term of the rectangular grid with pitch $\Delta x \times \Delta y$, the grid on the spatial frequencies is not evenly spaced, introducing a deformation of the Fourier transformed information [12]. To restore the computed amplitude to a linear representation, an additional interpolation step is performed which allows writing the spatial frequencies over a spherical grid. The core diffraction calculation using the Fresnel-Bluestein transform can be summarized as:

$$U = \mathcal{F}^{-1}\{\mathcal{F}\{f_1\} \cdot \mathcal{F}\{f_2\}\},$$

where,

$$\begin{aligned} f_1 &= S(m\Delta\xi, n\Delta\eta, z) \frac{\exp[ik|r_0|]}{r_0} \exp\left[\frac{ik}{2R}((\Delta\xi - \Delta x)\Delta\xi p^2 + (\Delta\eta - \Delta y)\Delta\eta q^2)\right] \\ f_2 &= \exp\left[\frac{ik}{2R}(\Delta x\Delta\xi m^2 + \Delta y\Delta\eta n^2)\right] \end{aligned} \quad (6)$$

Further details on the complete Fresnel-Bluestein expression can be found in [12]. Based on Fresnel diffraction, this method falls short on low propagation distances ($< \frac{\sqrt{2}}{2}M\Delta x$) and has been demonstrated to work for numerical apertures up to 0.58 [12].

4. Proposed digital lensless holographic modeling

To overcome the above-mentioned limitations in classical DLHM modeling methods, the proposed model separates the geometric and diffractive effects associated with the diverging-illumination free-space propagation and proposes alternative computational methods to generate them independently.

4.1. Magnification and geometrical distortion

Two critical geometrical aspects must be studied to accurately model the intensity distribution of the diffraction pattern captured by the sensor in DLHM. The first involves analyzing the magnification enabled by the DLHM geometry. The second examines the image distortion present in the recorded DLHM holograms, arising from the spherical nature of the illuminating wavefronts. Accounting for these factors is crucial for achieving a high-fidelity representation of the recorded intensity distributions. In the DLHM setup, assuming that a squared sensor is used, the system's effective numerical aperture (NA), as defined in Eq. (7), is determined by the sensor width W and the distance L when the entire sensor is illuminated. Essentially, this must

be smaller than the numerical aperture of the illuminating spherical wavefront:

$$NA = \frac{W/2}{\sqrt{(W/2)^2 + L^2}}. \quad (7)$$

On the other hand, the source-to-sample distance z , defines the field of view (FOV) captured by the system. Figure 3 visually illustrates how the area on the sensor's plane ($W \times W$) is geometrically projected to the corresponding area at the sample plane. The relationship between these two areas is determined by the overall magnification (Mag) of the DLHM system, which can be described by the ratio of the lengths of the arcs on the sample's ($arc_z : \theta z$) and sensor's wavefronts ($arc_L : \theta L$).

$$Mag = \frac{arc_z}{arc_L} = \frac{\theta L}{\theta z} = \frac{L}{z} \quad (8)$$

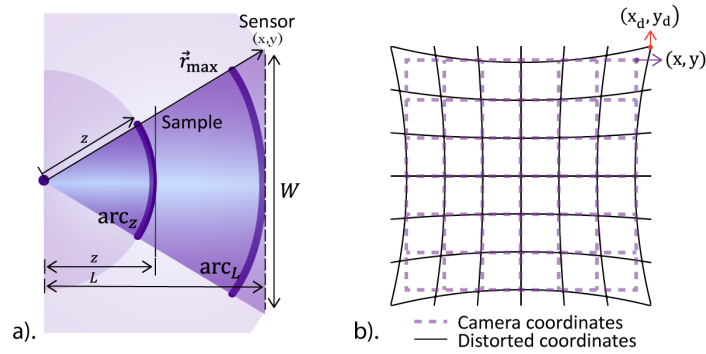


Fig. 3. (a) Scheme of the spherical wavefront on the sample and sensor's plane and (b) the recording of a spherical wavefront on a plane sensor yielding geometrical distortion.

While this magnification is physically associated with the diffraction pattern during the recording stage, the above description allows considering its effect as a geometrical remapping of the sample plane onto the sensor plane. Thus, before the diffractive effects are considered, the projection can be understood as a magnified mapping of the object [10]. According to geometrical optics, the projection S_{mag} and object S are related by,

$$S_{mag}(\xi, \eta, z) = \frac{1}{|Mag|} S\left(\frac{\xi}{Mag}, \frac{\eta}{Mag}, z\right). \quad (9)$$

The magnification Mag employed in Eq. (9) requires further study, considering the spherical illumination in DLHM. As depicted in Fig. 3(b), distortion is most pronounced at the edges of the DLHM recordings. This occurs because the source-to-sensor distance varies at each point, causing non-uniform magnification in the recorded hologram. Consequently, the projection of the sample becomes distorted in the DLHM hologram. This non-uniform magnification can be described as the ratio between the magnitude of the source-to-sensor vector $\vec{r}$ (see Fig. 1) and the source-to-sample distance, which is the radius of curvature z in Fig. 3(a). This leads to a definition of magnification as a function of the position in the sensor plane, as described by Eq. (10), where x and y represent the sensor coordinates. At the center of the sensor ($x = 0$ and $y = 0$), $\vec{r} = L$ and the magnification yields $Mag(0, 0) = L/z$.

$$Mag(x, y) = \frac{\sqrt{x^2 + y^2 + L^2}}{z} \quad (10)$$

The maximum distortion in the system occurs at the coordinates $(W/2, W/2)$, which correspond to the corners of the sensor. At this point, the maximum magnification Mag_{max} can be calculated

as indicated in Eq. (11)

$$Mag_{max} = \frac{\sqrt{\frac{W^2}{2} + L^2}}{z}. \quad (11)$$

These considerations can be extended for non-squared sensors by finding their maximum x and y coordinates and computing equations (10), and (11).

The varying magnification across the sensor distorts the recorded diffraction pattern; consequently, the rectilinear projection of the sample information must be likewise altered [30]. While distortion effects can be irregular in general, the use of spherical illumination sources in DLHM leads to radially symmetric distortions. Therefore, the maximum distortion in the DLHM system is computed as the difference between the magnification at the center and each coordinate, $|Mag_{max} - Mag|$. This proposal considers the distortion effect by interpolating the rectilinear projection of the sample on a spherical grid, similarly to what is proposed in the Fresnel-Bluestein transform [12]. To do so, the Brown-Conrady model, which is a well-known approach for radial distortion computation and correction [31], is utilized. The numerical implementation of this model is available in [32] and [33]. Using the first-order approximation of the model, the distorted coordinates can be expressed as

$$x_d = \frac{x + x_c K_1 r^2}{1 + K_1 r^2} \text{ and } y_d = \frac{y + y_c K_1 r^2}{1 + K_1 r^2} \quad (12)$$

where (x_d, y_d) are the distorted coordinates, (x, y) are the sensor's coordinates, which are the projection from an ideal spherical sensor, K_1 is the first radial distortion coefficient, which is computed considering the maximum magnification difference obtained in Eq. (11), (x_c, y_c) are the distortion center coordinates, and $r^2 = (x - x_c)^2 + (y - y_c)^2$.

In addition to the non-uniform magnification and image distortion modeled in this work, it is important to note that the variation in propagation distances across the sensor also introduces a defocus effect, particularly at higher NAs. While the current model implicitly accounts for some aspects of this phenomenon through distortion modeling, future work will explore more direct methods to compensate for defocus and further improve image reconstruction quality.

4.2. Diffraction

The diffraction effects observed in DLHM result from the propagation of spherical wavefronts as it impinges on the sample. One of the results of using spherical wavefronts as illumination is the magnification of the diffraction pattern by free-space propagation, as discussed earlier in subsection 4.1. This approach isolates this effect by employing a numerical resampling technique on the sample plane to model the said magnification properly. This process is depicted in panels (a) and (b) of Fig. 4. This resampling procedure ensures that the modified sample plane has the same physical size as the sensor plane, enabling the use of plane wavefront propagation. This change removes the discretization constraints of classical spherical wavefront propagation during the hologram modeling. In theory, any numerical diffraction method can be used, but considering that DLHM works on configurations with small distances that are in the order of millimeters, some of the propagators may fall short to adequately process the information. For these propagation requirements, the angular spectrum method (ASM) describes the requirements best [9,10]. The ASM considers the decomposition of the sample and hologram's field into their frequency components and relates one to the other by the free-space optical transfer function, given by the Kirchhoff-Helmholtz analysis. This propagation is modeled by a filtering process in Fourier domain, and it imposes that both planes have the same physical dimension when implemented using FFTs. Equations (13) and (14) show the convolution and the transfer function,

which acts as the propagation kernel, respectively.

$$U(p\Delta x, q\Delta y, L) = \mathcal{F}^{-1} \{ \mathcal{F} \{ S(m\Delta x, n\Delta y, z) \} \cdot H_z(f_x, f_y) \} \quad (13)$$

$$H_z(f_x, f_y) = \exp \left[i(L-z) \sqrt{\left(\frac{2\pi}{\lambda} \right)^2 - 4\pi^2((m\Delta f_x)^2 + (n\Delta f_y)^2)} \right] \quad (14)$$

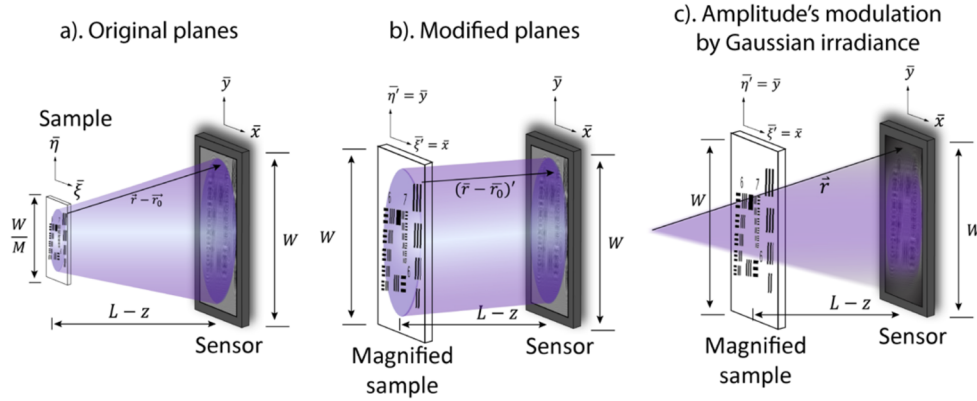


Fig. 4. Decomposition of the propagation process in DLHM. a) Physical magnification by sole propagation. b) Numerical magnification to mimic physical magnification of the sample information. c) Pointwise multiplication of the propagated sample wavefield by a spherical wavefront.

Here f_x and f_y are the corresponding spatial frequency components of the planes, whose discrete representation is given by $\Delta f_x = m/M\Delta x$ and $\Delta f_y = n/N\Delta y$. As the wavefield propagation is performed using the ASM, the computational efficiency of the proposed method is guaranteed since the algorithm only relies upon two FFTs. The ASM implementation used in this study is taken from [34].

Finally, to compensate for the uniform amplitude distribution caused by using a plane wavefront, the last step includes generating an amplitude modulation. The amplitude of the illumination wavefront is modeled as a Gaussian distribution, $|\text{PS}(\mathbf{r})| = \frac{\omega_0^2}{\omega(L)^2} \exp \left[-\frac{|\mathbf{r}|^2}{\omega(L)^2} \right]$, with $\mathbf{r}$ being the position vectors from the source the sensor plane, L the source-to-sensor distance, ω_0 the waist radius of the spherical wavefront source and $\omega(L)$ the radius at which the intensity falls to $1/e^2$ relative to the maximum intensity value in the recorded hologram. Mathematically, $\omega(L)$ is expressed $\omega(L) = \omega_0 \sqrt{1 + \left(\frac{L}{Z_R} \right)^2}$, where $Z_R = \pi \omega_0^2 n / \lambda$ is the Rayleigh range of the source and n represents the refractive index of the medium [35]. This amplitude is directly multiplied by the previously propagated optical wavefield at the sensor plane. The result is a modulation of the output's irradiance, as seen in panel (c) from Fig. 4. This same figure provides a graphical representation of the entire process.

5. Algorithm description

The working principle of the model, the magnification of the sample recorded by the sensor, and the associated distortion involved in the light propagation of DLHM have been outlined above. In this section, these principles are combined into an algorithm that enables the generation of modeled DLHM holograms through the five steps described below. The algorithm is available at the public repository [36] and a flowchart is provided in the [Supplement 1](#).

- i) Resampling of the object plane:** to achieve the numerical magnification described in section 4.1, a portion of the object plane is cropped. This area is selected so that it meets a physical size of $\frac{P \cdot \Delta x}{Mag} \times \frac{Q \cdot \Delta y}{Mag}$. This resampling procedure eliminates the constraints on the number of pixels at the input sample plane and its physical dimensions. This allows for the simulation of holograms of objects of any size with adjustable discretization. If the sample's discretization $\Delta \xi \times \Delta \eta$ is not provided, the algorithm assumes that the sample plane has a discretization pixel given by $\Delta \xi \times \Delta \eta = \frac{\Delta x}{Mag} \times \frac{\Delta y}{Mag}$. Then, the sample plane, which is discretized on a $P \times Q$ grid must be resampled to have the same discretization of the sensor's plane $M \times N$. This process is performed by a digital image rescaling. The result is that both the sample and the sensor plane have the same numerical and physical size.
- ii) Plane wavefront propagation of the resampled object information and hologram computation:** the resampled object is propagated a total $L - z$ distance with the angular spectrum formalism, assuming a plane wave illumination. Then, the intensity of the field at the sensor plane is computed as stated by Eq. (2).
- iii) Interpolation for spheric coordinates (geometrical distortion):** as described in section 4.1, the recording on a plane sensor induces a radial distortion on the resulting image. This distortion is induced in this step by using the first-order approximation of the Brown-Conrady model, using as the K_1 parameter the maximum distortion encountered by the difference between the magnification at the center and the edges of the sensor (section 4.1).
- iv) Superposition with Gaussian irradiance:** the amplitude of the point source illumination $|PS(r)|$ is computed independently as a Gaussian distribution, and a pointwise multiplication between this irradiance and the magnified and propagated information of the object is performed.
- v) Hologram normalization:** to better represent the realistic experimental recording, an 8-bit normalization is performed.

6. Ranges of validity: comparison among all the methods

The main aspect that limits the use of numerical propagators to model diffraction phenomena is their capacity to properly work according to the propagation distance, and number and size of discretization pixels, in both sample and detection planes [9]. Due to the approximations and assumptions inherent in each method, discretization plays a pivotal role in determining the propagation range in which each method works with minimum error. This is particularly relevant for modeling DLHM holograms close to experimental conditions, where discretization is primarily dictated by the sensor characteristics. To compare this crucial feature among the various discussed simulation methods, including the proposed model, a series of simulated experiments were conducted. In these experiments, holograms of a star-shaped sample with internal Siemens star frequency information were simulated under varying numerical apertures, magnifications, and pixel pitches. The original sample is shown in panel (a) of Fig. 5. The numerical apertures and magnifications were selected within the common operation range of DLHM systems, while the pixel pitch was considered for three cases, based on common commercially available sensors. Table 1 summarizes the selected ranges for each variable.

As DLHM systems commonly operate in the near-field diffraction regime, the lower spatial frequencies from the holograms produced by any simulation method should preserve an overall resemblance to the original object in the regions that would constitute a geometric shadow. This property can thus be harnessed to estimate whether each method is capable of simulating valid holograms for a given configuration as follows. After obtaining the output hologram, an example

Table 1. Variables used to simulate holograms under diverse conditions with different propagation models.

Variable	Range
DLHM numerical aperture	[0.05–0.8]
Magnification	[1.1–20] ×
Sampling (pixel pitch and # of pixels)	1.85 μm (3000 × 3000), 2.40 μm (3648 × 3648), 3.60 μm (3500 × 3500)

of which is provided on panel (b), it is resized into a grid of 1% the size of the original image, as illustrated in panel (f). This rescaling induces an undersampling of the diffraction pattern, preserving only the low-frequency content. The same process is performed for the original sample, as presented in panel (e). These two images, panels (e) and (f), are then compared by computing their correlation coefficient. As visualized, when a simulated hologram does not suffer from any numerical artifacts that significantly distort the output, the resized version closely resembles the original object and produces a high correlation value. Conversely, when the propagators fail and produce invalid holograms, as those shown in panels (c) and (d), the correlation drops, as seen in panels (g) and (h). Consequently, the validity of a method for each given configuration is determined by the correlation metric, where values larger than 0.5 are considered valid.

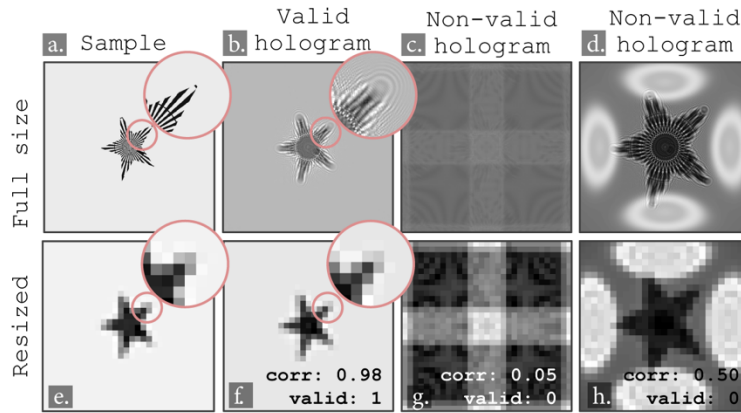


Fig. 5. Simulation to evaluate the ranges of validity of the modeling methods (a) Star target used as input. (b) a valid hologram. (c) non-valid hologram and (d) non-valid hologram with a correlation value of 0.50. Panels (e), (f), (g) and (h) showcase the resized images and their correlation value to the input sample. The zoom in areas highlight the reduction of the high frequency content of the holograms after the undersampling procedure.

A map of all the correlation values obtained from these experiments is displayed in Fig. 6. The propagation distances L were calculated for each sensor configuration using Eq. (7), based on their known pixel sizes and widths. For each L , the corresponding sample-to-source distances z were determined using Eq. (8) for magnifications ranging from 1.1× to 20×.

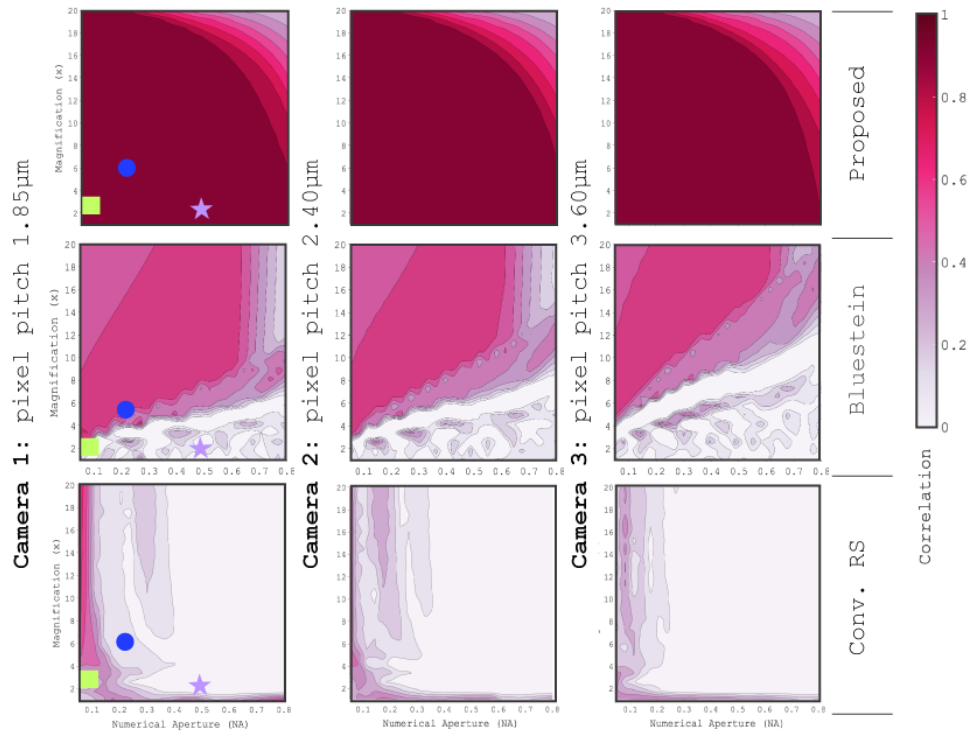


Fig. 6. Range of validity of the DLHM simulation methods in terms of numerical aperture, magnification and pixel pitch. The colored markers are the selected configurations for the experimental validation presented in section 6.

7. Experimental validation

Based on the results provided by Fig. 6, three validation scenarios are selected, and presented with the colored markers (green square, blue circle and purple star). The validation consists of experimentally acquiring holograms under each recording configuration and comparing the result with what can be simulated using each model. The sample to study is a calibrated phase target from Benchmark Technologies. With the same configurations used for the experimental recording, holograms are simulated using the proposed methods. The non-approximated Rayleigh-Sommerfeld (Discrete RS) method is also included for comparison. As this method does not account for any approximations, both planes can be sampled accordingly to achieve a valid hologram. For further information about the sampling requirements, refer to [9]. The simulations use a digital model of the phase target as input. The geometrical configurations of the three scenarios with their parameters are displayed in Table 2, these geometrical parameters were calibrated via the method described in [37]. For all scenarios, the sensor features a pixel size of $1.85\ \mu\text{m}$ and 3000×3000 pixels, resulting in an effective sensor's width of $5.56\ \text{mm} \times 5.56\ \text{mm}$. All the experiments are performed with a $532.8\ \text{nm}$ laser.

Figure 7 illustrates the holograms obtained for the first configuration using all the methods. The white dashed rectangle highlights the area of interest, which is reconstructed and presented below each hologram. All the reconstructions are performed using the angular spectrum algorithm with convergent spherical wavefront illumination, taking into account the necessary sampling criteria to prevent aliasing effects [10]. The reconstruction algorithm is available in [36]. The reconstructions focus on a Siemens Star object. On the bottom panel of each hologram, a line plot is displayed. This line is taken over the position in which the minimum-resolvable circumference

Table 2. Description of the geometrical parameters of the experimental holograms.

	Marker	L (mm)	z (mm)	Mag	NA	Valid methods
Configuration 1		31.45	8.20	3.8×	0.09	All methods
Configuration 2		17.04	3.00	5.7×	0.16	Proposed, Fresnel-Bluestein
Configuration 3		5.90	5.30	1.1×	0.42	Proposed only

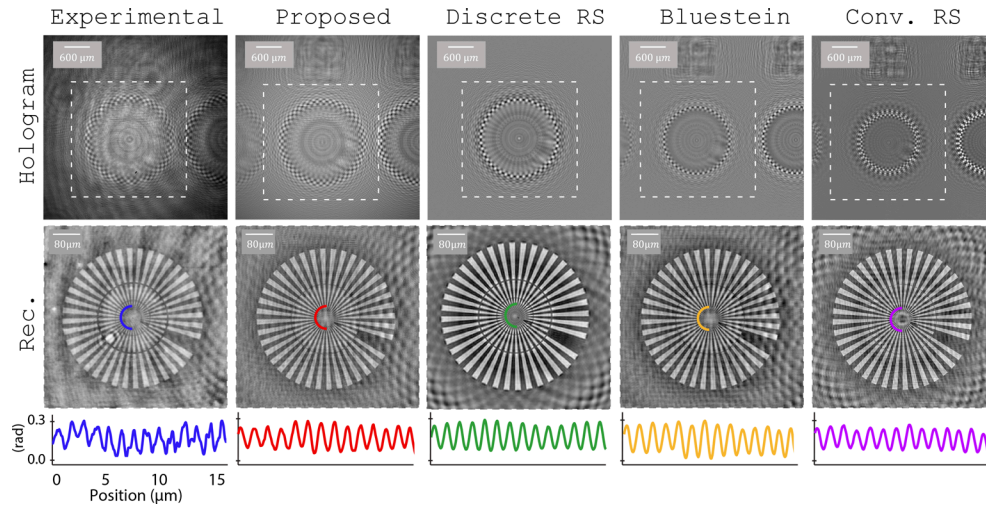


Fig. 7. Results of Configuration 1 showing the experimental and simulated holograms with the proposed model, the discrete RS, the Bluestein transform, and the convolutional RS. For each hologram the reconstruction of a Siemens Star is provided on the black square. The crossed section of the reconstruction at the maximum achieved resolution is presented on the bottom of each panel.

is found for the experimental result. This circumference defines the lateral resolution of the system, which corresponds to $2.4 \mu\text{m}$. The reconstructed profiles show a similar behavior across the methods, the changes presented are due to the twin image effect that varies among the methods, and the noise coming from the experimental setup. In general, all reconstructions achieve a resolution comparable to the experimental results. This consistency across methods confirms that under the conditions of Configuration 1, each model can reliably simulate the holograms.

Figure 8 displays the results obtained for the second configuration. In this case, the convolutional RS method fails to provide a valid hologram, as can be appreciated artifacts decrease the reconstruction quality and overall do not resemble experimental outcomes. The experimental hologram showcases that the maximum resolvable object is the element 7-3 from the USAF 1951 test target, corresponding to a resolution of $3.10 \mu\text{m}$. The simulated holograms using the proposed, the discrete RS and the Fresnel-Bluestein methods exhibit similar behavior, correctly identifying the maximum resolvable frequency. These results confirm the validity of the methods under this configuration, aligning well with the experimental observations.

The last result, presented in Fig. 9, shows the third configuration, in which only the proposed model and the non-approximated Rayleigh-Sommerfeld (Discrete RS) can provide a valid hologram. Only the area enclosed by the purple square was computed using the Discrete RS method, as this is the region required for analysis, and to minimize processing time. The failed holograms are also displayed, highlighting how in this case the results do not contain any valid

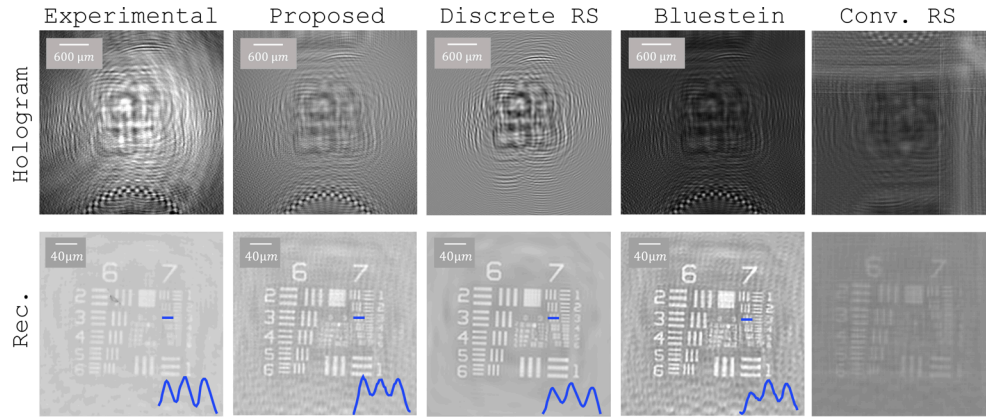


Fig. 8. Results of Configuration 2 showing the experimental and simulated holograms with the proposed model, the discrete RS, the Bluestein transform, and the convolutional RS. The crossed section of all the reconstructions at the maximum achieved resolution is displayed at the right border of each reconstructed image.

information. This configuration uses a very low magnification and a large field of view, as seen on the obtained holograms. In this case, the resolution is mainly determined by the sensor's pixel pitch; namely, $1.85 \mu\text{m}$, which corresponds to the element 8-1 of the USAF target, the profile of the element 8-2 of the target is also presented in the blue lines. The purple zoom-in areas highlight the portion of the hologram for which the reconstruction is performed and shown, due to the high sampling required for the non-approximated Rayleigh-Sommerfeld, only this portion was propagated for this case. From this result, the remarkable similarity between the resulting reconstructions must be highlighted: both the object features and the alterations attributable to the presence of the twin image in the experimental result are clearly and accurately modeled by the proposed simulation method. This is further shown as the achieved spatial resolution is the same for the three results.

One key aspect to emphasize regarding the methods is their computational complexity. Assuming all images consist of M -by- M pixels, Table 3 outlines the computational complexity for each algorithm. The performance of these algorithms can be evaluated indirectly through their overall processing times. Holograms with dimensions of 3000 by 3000 pixels were simulated using the presented techniques. For the proposed method, the Convolutional Rayleigh-Sommerfeld, and the Bluestein methods, 50 holograms were computed, and the average time is displayed in Table 3. Moreover, due to its longer processing time, the Discrete Rayleigh-Sommerfeld method was run five times, and the average time is calculated and presented in Table 3. All tests were conducted on a computer equipped with an Intel Core i7-8750 H CPU and a NVIDIA GeForce GTX 1050 Ti with Max-Q Design. The results highlight the computational advantages of the proposed method, which not only delivers accurate hologram simulations but also does so with remarkable speed compared to the other methods.

Table 3. Computational complexity and average running time of the simulation methods.

	Proposed	Discrete RS	Bluestein	Convolutional RS
Computational complexity (O)	$10 M^2 + 2 M^2 \log(M^2)$	M^4 [9]	$10 M^2 + 3 M^2 \log(M^2)$ [38]	$3 M^2 \log(M^2)$
Time (seconds)	0.99 ± 0.06	52330 ± 30	7.6 ± 0.4	2.02 ± 0.08

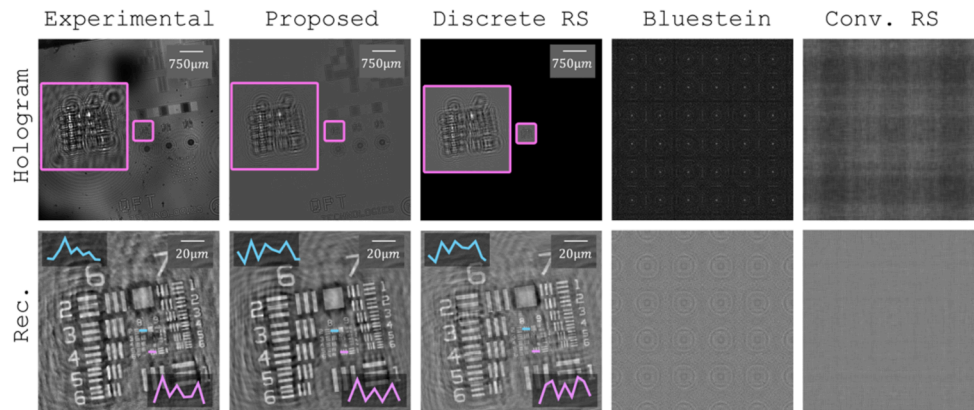


Fig. 9. Results of Configuration 3 showing the experimental and simulated holograms with the proposed model and the discrete RS, the Bluestein transform, and the convolutional RS results are also displayed showcasing how these methods failed to provide a proper hologram. For each valid hologram a zoom in area is presented in pink rectangle together with the reconstruction of a USAF 1951 target. The crossed section of elements 8-1 and 8-2 of all the reconstructions is displayed at the bottom right corner and upper left corner of each reconstructed image respectively.

8. Conclusions

This work presents a model for computational simulation of DLHM holograms by decomposing the various physical and geometrical phenomena intrinsic to the technique. Each phenomenon is individually addressed and integrated into an algorithm that produces accurate holograms. The model serves as an open access [36] cutting-edge tool for simulating DLHM holograms, providing both computational efficiency and high-fidelity holograms compared to current state-of-the-art approaches. The validity of the model is confirmed through rigorous testing, where simulated holograms are compared with experimental holograms of standard resolution test targets, including USAF 1951 and Siemens Star layouts. This comparative analysis also includes other DLHM hologram modeling techniques, such as the Rayleigh-Sommerfeld diffraction integral, its convolutional version, and the Fresnel-Bluestein method. The applicability range of these methods is initially heuristically mapped for numerical apertures between 0.05 and 0.8 and magnifications ranging from $1.1\times$ to $20\times$, allowing the selection of critical regions where the existing simulation approaches fail.

The proposed model demonstrates superiority over established methods by consistently generating reliable holograms across the tested regions, with reconstructions that accurately reflect experimental results. Additionally, the proposed model features lower computational complexity, resulting in a more efficient algorithm in terms of processing times. This efficiency is particularly valuable for applications requiring real-time, high-throughput hologram modeling and for experimental settings. These advancements contribute to the progression of DLHM and lay the groundwork for future developments in learning-based methods and data-efficient hologram processing techniques, ensuring the continued evolution of this technology. As future work, a modified version of this approach is being developed to include DLHM hologram reconstruction, thereby enhancing its applicability and impact in the field.

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Data availability. The model is available at the public repository [36].

Supplemental document. See [Supplement 1](#) for supporting content.

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